MASONRY:

1. THE MASONRY PORTIONS OF THIS STRUCTURE ARE DESIGNED ACCORDING TO THE LATEST WORKING STRESS DESIGN PROVISIONS OF THE MASONRY STANDARDS JOINT COMMITTEE (MSJC) BUILDING CODE REQUIREMENTS FOR MASONRY STRUCTURES (ACI 530/ASCE 5/TMS 402) AND SPECIFICATIONS FOR MASONRY STRUCTURES (ACI 530.1/ASCE 6/TMS 602) INCLUDING SECTIONS 2106 AND 2107 OF CHAPTER 21 IN THE MICHIGAN BUILDING CODE. MASONRY COMPONENTS HAVE BEEN DESIGNED ACCORDING TO THE PROVISIONS FOR SEISMIC DESIGN CATEGORY B.
2. ALL STRUCTURAL MASONRY IS TO BE CONSTRUCTED IN ACCORDANCE WITH THE LATEST MASONRY STANDARDS JOINT COMMITTEE (MSJC) BUILDING CODE REQUIREMENTS FOR MASONRY STRUCTURES (ACI 530.1/ASCE 6/TMS 602). MASONRY SUBMITTALS ARE REQUIRED BY ACI 530.1/ASCE 6/TMS 602 SECTION 1.6, TABLE 1.15.2, LEVEL B, LEVEL C QUALITY ASSURANCE.
3. ALL STRUCTURAL MASONRY HAS BEEN ENGINEERED IN ACCORDANCE WITH CHAPTER 2, ALLOWABLE STRENGTH DESIGN. COMPRESSION STRENGTH SHALL BE DETERMINED ACCORDING TO THE UNIT STRENGTH METHOD FOR CONCRETE MASONRY MSJC SECTION 1.4.B.2.b.
4. ALL CONCRETE MASONRY UNITS SHALL CONFORM TO ASTM C 90 AND C 140.
5. MORTAR SHALL CONFORM TO ASTM C 270, AS FOLLOWS:
   - IN CONTACT WITH EARTH - TYPE M
   - NOT IN CONTACT WITH EARTH - TYPE M OR S
   - INTERIOR, NON-LOAD BEARING PARTITIONS - TYPE N
   - VENEERS - TYPE N IS ACCEPTABLE IF COORDINATED WITH OTHER ARCHITECTURAL REQUIREMENTS
6. GROUT SHALL CONFORM TO ASTM C 476, WITH PEA GRAVEL AGGREGATE AND WITH MINIMUM STRENGTH IN ACCORDANCE WITH THE TABLE BELOW.
7. BASED ON THE FOLLOWING MINIMUM STRENGTHS SPECIFIED ABOVE FOR BLOCK, MORTAR AND GROUT, THE MINIMUM COMPOSITE MASONRY COMPRESSIVE STRENGTH, f3m, SHALL BE AS FOLLOWS:

USE	COMPRESSIVE STRENGTH, f3m (PSI)	UNIT STRENGTH (PSI)	GROUT STRENGTH (PSI)	MOTAR TYPE
TYPICAL U.N.O.	2000 MIN.	2800 MIN.	2000 MIN.	M OR S

8. THE FOLLOWING LAP SPLICE REQUIREMENTS SHALL APPLY:

WALL CONFIGURATION	#4 BAR	#5 BAR	#6 BAR	#7 BAR	#8 BAR
8" WALLS (SINGLE REINFORCEMENT)	24"	30"	36"	--	--
8" WALLS (DOUBLE REINFORCEMENT)	24"	30"	58"	--	--
12" WALLS (SINGLE REINFORCEMENT)	24"	30"	36"	42"	48"
12" WALLS (DOUBLE REINFORCEMENT)	24"	30"	58"	80"	--

9. ALL STRUCTURAL MASONRY SHALL COURSE IN STANDARD RUNNING BOND. UNLESS NOTED OTHERWISE, ALL INTERSECTING BEARING WALLS, SHEAR WALLS OR OTHER STRUCTURAL WALLS SHALL BE LAID UP IN INTERLOCKED, BONDED COURSING. MECHANICAL ANCHORS OR WALL TIES MAY BE SUBSTITUTED WITH PRIOR APPROVAL BY THE ENGINEER.
10. PROVIDE HORIZONTAL WIRE TYPE REINFORCING WITH 9 GAUGE SIDE RODS AND 9 GAUGE CROSS RODS IN EVERY SECOND COURSE (16" O.C.) IN ALL MASONRY WALLS. SPACE AT 8" O.C. AT PARAPET WALLS. PROVIDE "LADDER" TYPE REINFORCING ONLY IN WALLS WITH VERTICAL REINFORCING. PROVIDE ADJUSTABLE TIES AT ALL LINTELS AND CAVITY WALLS AT 16" O.C. MAXIMUM SPACING.
11. ALL REINFORCING BARS, DOWELS AND TIES SHALL CONFORM TO ASTM A615, GRADE 60. VERTICAL REINFORCING BARS SHALL BE HELD IN PLACE BY POSITIONERS. SEE SPECIFICATION FOR SPACING.
12. PROVIDE A CONTINUOUS BOND BEAM, WITH 2-#5'S. AT THE TOP OF WALLS PARALLEL WITH ROOF/FLOOR FRAMING. STEP BOND BEAMS ELEVATIONS AS REQUIRED, LAP MINIMUM 32".
13. PERFORM GROUTING ACCORDING TO THE FOLLOWING:
   - ALL MASONRY BELOW GRADE SHALL BE GROUTED SOLID.
   - ALL CORES WITH VERTICAL REINFORCING OR TO RECEIVE DRILLED-IN ANCHORS SHALL BE GROUTED SOLID.
   - ALL MASONRY PIERS AND PILASTERS SHALL BE GROUTED SOLID.
   - MAXIMUM 5'-0" HIGH POUR HEIGHT.
14. ALL BEAMS SUPPORTING MASONRY, INCLUDING STEEL, PRECAST AND MASONRY LINTELS ARE TO BEAR 8" (MIN.) ON 3 COURSES SOLID MASONRY.

LUMBER:

1. ALL LUMBER WORK SHALL BE IN ACCORDANCE TO 2015 NATIONAL DESIGN SPECIFICATION (NDS) AND 2015 MICHIGAN BUILDING CODE.
2. ALL 2X WOOD STUDS, JOISTS, AND POSTS SHALL BE SOUTHERN PINE, No. 2 (OR BETTER) AND KILN-DRIED WITH THE FOLLOWING MINIMUM PROPERTIES:
   a. Fb = 750 PSI
   b. FV = 175 PSI
   c. FC || = 1,250 PSI
   d. E = 1,400,000 PSI
3. RAFTERS, JOISTS AND HEADERS SHALL BEAR ON A MINIMUM OF 3 ½" WIDTH.
4. ALL METAL CONNECTIONS SHALL BE SIMPSON STRONG TIE.
5. ALL FASTENING REQUIREMENTS SHALL BE IN ACCORDANCE WITH THE APPLICABLE PROVISIONS OF SECTION 2304.10 AND TABLE 2304.10.1.
6. BUILT-UP HEADERS SHALL BE ATTACHED WITH CONSTRUCTION GLUE AND GRK SCREWS AT 10" O.C. STAGGERED TOP + BOTTOM. HEADERS SHALL HAVE THE FOLLOWING CONSTRUCTION (REFER TO PLAN FOR HEADER DEPTHS):
   A. 2X4 WALL: (2) 2x10 (MIN.) + 1/2" PLYWOOD (3 1/2" TOTAL WIDTH)
   B. 2X6 WALL: (3) 2x10 (MIN.) + (2) 1/2" PLYWOOD (5 1/2" TOTAL WIDTH)
   C. 2X8 WALL: (4) 2x10 (MIN.) + (3) 3/8" PLYWOOD (7 1/8" TOTAL WIDTH)
7. MULTI-PLY BUILT-UP WOOD OR LVL BEAMS SHALL BE ATTACHED FOR FULL LAMINATION USING CONSTRUCTION GLUE AND 5/16" DIA. GRK SCREWS AT 10" O.C. STAGGERED TOP AND BOTTOM BETWEEN EACH ADJACENT PAIR OF PLIES.
8. WHERE FLOOR JOISTS ARE PARALLEL WITH BEARING, PROVIDE 2X SOLID BLOCKING AT 16" O.C. AT FIRST BAY (LADDER BLOCKING).
9. ALL HEADERS SHALL HAVE A MINIMUM (2) 2X JACK STUDS AT BEARING ENDS (REFER TO PLANS). PROVIDE MINIMUM (2) 2X KING STUDS AT BEARING ENDS.
10. POINT LOAD ABOVE (PLA) AS SHOWN ON PLAN MUST BE SUPPORTED WITH AT LEAST (2) 2X STUDS WITHIN THE 2X WALL. ALL POINT LOADS ABOVE MUST BE SUPPORTED WITH A HEADER, BEAM OR MULTIPLE STUDS.
11. BALLOON FRAMING STUDS SHALL BE 2X6 MINIMUM. PROVIDE (2) 2X JACK STUDS AT BEARING ENDS AND (2) KING STUDS MINIMUM.
12. ALL TJI RAFTERS AND JOISTS SHALL BE BY TRUS JOIST. REFER TO INSTALLATION RECOMMENDATIONS FROM TRUS JOIST FOR TYPICAL FRAMING DETAILS.
13. ENGINEERED LUMBER INCLUDE LAMINATED VENEER LUMBER (LVL), PARALLEL STRAND LUMBER (PSL) AND LAMINATED STRAND LUMBER (LSL) WITH THE FOLLOWING PROPERTIES:
   G. LVL:
      a. E = 1,900,000 PSI MIN.
      b. WIDTH: 1 ¾"
      c. DEPTHS: 5 ½", 7 ¼", 9 ¼", 9 ½", 11 ¼", 11 7/8", 14", 16", 18", 20" AND 24"
   B. PSL:
      a. E = 2,000,000 PSI
      b. WIDTHS: 3 ½", 5 ¼" AND 7"
      c. DEPTHS: 9 ¼", 9 ½", 11 ¼", 11 7/8", 14", 16" AND 18"
   C. LSL:
      a. E = 1,550,000 PSI
      b. WIDTHS: 1 ¾" AND 3 ½"
      c. DEPTHS: 9 ¼", 9 ½", 11 ¼", 11 7/8", 14" AND 16"

   APPROVED ENGINEERED LUMBER MUST BE FROM THE FOLLOWING:
      a. TRUS JOIST
      b. BOISE CASCADE
      c. GEORGIA PACIFIC
      d. LOUISIANA PACIFIC
      e. RED BUILT
14. DO NOT DRILL HOLES, NOTCH OR CUT EXISTING LUMBER FRAMING WITHOUT CONSULTING THE ARCHITECT OR ENGINEER OF RECORD.
15. CONTRACTOR IS RESPONSIBLE FOR PROVIDING TEMPORARY SHORING BEFORE AND DURING LUMBER FRAMING INSTALLATION.
16. ROOF SHEATHING SHALL BE 3/8" APA RATED PLYWOOD. ATTACHED TO WOOD FRAMING WITH CONSTRUCTION GLUE AND 8d NAILING @ 6" O.C. EDGE AND 12" O.C. FIELD NAILING.
17. EXTERIOR WALL SHEATHING SHALL BE 1/2" APA RATED PLYWOOD. ATTACH TO WOOD FRAMING WITH 8d @ 6" O.C. EDGE AND 12" O.C. FIELD NAILING. REFER TO SHEAR WALL FRAMING FOR SHEAR WALL NAILING IF REQUIRED.
18. FLOOR SHEATHING SHALL BE 23/32" APA RATED TONGUE AND GROOVE PLYWOOD. ATTACH TO WOOD FRAMING WITH CONSTRUCTION GLUE AND 2" #8 WOOD SCREWS (OR 8d NAIL) AT 6" O.C. EDGE AND 12" O.C. FIELD NAILING.

FOUNDATIONS:

EXCAVATIONS

1. FOOTINGS SHALL BEAR ON FIRM, UNDISTURBED SOIL OR ENGINEERED FILL WITH AN ASSUMED SAFE BEARING CAPACITY OF 2000 PSF. IF SOIL OF THIS CAPACITY IS NOT FOUND AT THE ELEVATIONS INDICATED, FOOTINGS SHALL BE ENLARGED OR LOWERED AT THE DIRECTION OF THE ARCHITECT. ALLOWABLE SOIL BEARING PRESSURE SHALL BE CONFIRMED IN THE FIELD BY A QUALIFIED SOILS ENGINEER.
2. UNLESS OTHERWISE NOTED OR DETAILED, ALL FOUNDATIONS SHALL BE LOCATED SUCH THAT THE CENTERLINE OF FOOTING IS ALSO THE CENTERLINE OF COLUMN.
3. BOTTOM OF ALL EXTERIOR FOOTINGS MUST BE 3'-6" MIN. BELOW GRADE.
4. COORDINATE ALL EMBEDDED ITEMS, SLEEVES THRU-FOOTING, OPENINGS AND ELECTRICAL CONDUITS WITH MECHANICAL, CIVIL, STRUCTURAL AND ARCHITECTURAL DRAWINGS PRIOR TO PLACEMENT OF CONCRETE.
5. PROVIDE NECESSARY SHEETING, SHORING, FORMING OR BRACING, ETC., DURING EXCAVATION AS REQUIRED TO PROTECT SIDES OF EXCAVATIONS OR AS REQUIRED TO COMPLY WITH SAFETY REGULATIONS. DO NOT BACKFILL BEHIND BASEMENT WALLS UNTIL FLOOR FRAMING OR TEMPORARY BRACING IS IN PLACE.
6. REFER TO THE PROJECT GEOTECHNICAL REPORT FOR UNDERCUTTING OF EXISTING POOR QUALITY FILL AND BURIED TOPSOIL WHICH MUST BE REMOVED FROM WITHIN THE AREA OF THE BUILDING ADDITION INCLUDING 10 FEET BEYOND THE BUILDING ADDITION FOOTPRINT. PROVIDE ENGINEERED BACKFILL TO THE UNDERSIDE OF THE NEW FLOOR SLAB. REFER TO PROJECT GEOTECHNICAL REPORT FOR UNACCEPTABLE FILL DISPOSAL GUIDELINES.
7. PREPARATION OF THE SITE, BUILDING FOOTPRINT AND SLAB SUB-BASE SHALL PROCEED IN COMPLIANCE WITH THE LOCAL CODES AND THE PROJECT SOILS REPORT IDENTIFIED ABOVE. UNLESS OTHERWISE NOTED OR SPECIFIED., ALL FILL UNDER FLOOR SLABS AND BEHIND FOUNDATION WALLS SHALL BE COMPACTED WITH VIBRATORS, COMPACTORS, ETC. TO 95% MAXIMUM DENSITY (MODIFIED PROCTOR) AT OPTIMUM MOISTURE CONTENT.
8. THIS TRADE SHALL PROVIDE PUMPS, WELL POINTS, OR OTHER SYSTEMS AS REQUIRED BY THE CONDITIONS IDENTIFIED IN THE PROJECT SOILS REPORT. PUMPS SHALL BE OPERATED AS REQUIRED TO ACCOMPLISH THE ABOVE, ON A 24-HOUR BASIS, IF NECESSARY. UNDER NO CONDITIONS SHALL WATER BE ALLOWED TO WASH OVER FRESHLY PLACED CONCRETE.

CONCRETE:

1. THE CONCRETE PORTIONS OF THIS STRUCTURE ARE DESIGNED ACCORDING TO THE LATEST ULTIMATE STRENGTH DESIGN PROVISIONS OF THE AMERICAN CONCRETE INSTITUTE BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE AND COMMENTARY (ACI 318) INCLUDING SECTIONS 1902 THRU 1907 OF CHAPTER 19 IN THE MICHIGAN BUILDING CODE. CONCRETE COMPONENTS HAVE BEEN DESIGNED ACCORDING TO THE PROVISIONS FOR SEISMIC DESIGN CATEGORY B.
2. ALL CONCRETE SHALL BE NORMAL WEIGHT (150 PCF). MINIMUM CONCRETE STRENGTH SHALL BE f 'c = 3000 PSI MIN. AT 28 DAYS, UNLESS NOTED OTHERWISE; SUPPORTED SLABS AND SLABS ON GRADE SHALL BE f 'c = 3000 PSI MIN. UNLESS NOTED OTHERWISE. PROVIDE f 'c = 4000 PSI WITH 6% +/- 1% ENTRAINED AIR WHERE CONCRETE IS EXPOSED TO EXTERIOR ATMOSPHERE OR WEATHER.
3. ALL CONCRETE SHALL BE PORTLAND CEMENT CONFORMING TO ASTM C150. AGGREGATE SHALL CONFORM TO ASTM C33.
4. CONCRETE ADMIXTURES SHALL BE USED TO FACILITATE CONCRETE PLACEMENT, AID DIFFICULT PLACING CONDITIONS OR ASSIST IN ATTAINING SPECIFIED CONCRETE QUALITIES. ADMIXTURES SHALL HAVE LESS THAN 0.05 CHLORIDE IONS.
   - AIR ENTRAINMENT PER ASTM C260
   - WATER REDUCER PER ASTM C494, TYPE A
   - WATER REDUCER / ACCELERATOR PER ASTM C494, TYPE C OR E
   - WATER REDUCER / RETARDER PER ASTM C494, TYPE B OR D
   - SUPERPLASTICIZER PER ASTM C494, TYPE F OR G
5. CONCRETE MIXES SHALL BE PROPORTIONED PER SECTION 3.9 OF ACI-301. CERTIFIED HISTORICAL TEST DATA SHALL SERVE AS A BASIS FOR EACH MIX DESIGN. WHERE HISTORICAL TEST DATA IS NON-EXISTENT THE FOLLOWING GUIDELINES SHALL APPLY:

	COMPRESSIVE STRENGTH, f 'c	CEMENT CONTENT	WATER / CEMENT RATIO	
TYPE	(28 DAYS, PSI)	(LBS. / C.Y.)	(BY WEIGHT)	SLUMP
STANDARD, NORMAL WT.	3000 MIN.	470 MIN.	0.55 MAX.	4" MIN.
STANDARD, NORMAL WT.	3500 MIN.	517 MIN.	0.50 MAX.	4" MIN.
AIR ENTRAINED, NORM. WT.	4000 MIN.	564 MIN.	0.45 MAX.	4" MIN.

6. ALL CONCRETE WORK AND PLACEMENT SHALL CONFORM TO THE LATEST ACI STANDARDS AND RECOMMENDATIONS. FREE FALL SHALL NOT EXCEED 10 FEET FOR ALL CONCRETE CONTAINING HIGH-RANGE WATER REDUCER (SUPERPLASTICIZER) AND 5 FEET FOR ALL OTHER CONCRETE. PROVIDE ELEPHANT TRUNK, TREMIES OR OTHER PLACING EQUIPMENT OR OPENINGS IN SIDES OF FORMS AS REQUIRED TO LIMIT FREE FALL.
7. ALL REINFORCING BARS, DOWELS AND TIES SHALL CONFORM TO ASTM A615, GRADE 60. ALL REINFORCING STEEL SHALL BE CONTINUOUS AND SHALL HAVE 36 BAR DIAMETER LAP (MIN.) AND SHALL BE FABRICATED AND PLACED IN ACCORDANCE WITH ACI 315 AND ACI 318, LATEST EDITION. HOOK ALL BEAM BARS AT DISCONTINUOUS ENDS.
8. ALL WELDED WIRE FABRIC SHALL CONFORM TO ASTM A185 FURNISHED IN FLAT MATS OR SHEETS, NOT IN ROLLS. PROVIDE MINIMUM 8" LAP BETWEEN SHEETS. ALL SLAB REINFORCING SHALL BE SUPPORTED ON SAND CHAIRS.
9. ALL COLUMN / PIER OR WALL DOWELS SHALL BE 48 DIAMETERS LONG USING THE SAME SIZE AND QUANTITY AS THE VERTICAL COLUMN / PIER OR WALL BARS. PROVIDE HOOKS AS REQUIRED FOR PLACEMENT.
10. ALL EXPOSED CONCRETE CORNERS AND EDGES SHALL BE CHAMFERED 3/4".
11. POURED CONCRETE WALLS SHALL BE VIBRATED IN 20" MAX. LIFTS DURING THE PLACEMENT OPERATION.
12. MINIMUM CONCRETE COVERS SHALL BE (UNLESS OTHERWISE NOTED):
   A. UNFORMED SURFACES IN CONTACT WITH GROUND (FOOTING BOTTOMS). 3"
   B. SLABS ON GRADE (TOP COVER). 1"
   C. FORMED SURFACES IN CONTACT WITH GROUND OR EXPOSED TO THE WEATHER (GRADE BEAMS, WALLS, ETC.) 2"
   D. BEAMS AND COLUMN MAIN REINFORCING OR STIRRUPS AND TIES $1\frac{1}{2}$"
   E. INTERIOR & EXTERIOR STRUCTURAL TIES 1"
   F. INTERIOR WALL SURFACES 1"
   G. IN ALL CASES, CLEARANCE NOT LESS THAN THE DIAMETER OF BARS.

NOTE: MAXIMUM DEVIATION FROM THESE REQUIREMENTS SHALL BE +1/4" FOR SECTIONS TEN (10) INCHES OR LESS AND +1/2" FOR SECTIONS OVER TEN (10) INCHES THICK.

GENERAL NOTES:

GENERAL CONDITIONS

1. REFER TO ARCHITECTURAL DRAWINGS FOR OVERALL DIMENSIONS, LAYOUT, WALL HEIGHTS, FINISH FLOOR ELEVATIONS, GRADING AND OTHER TRADES.
2. IF ANY GENERAL NOTE CONFLICTS WITH ANY DETAIL OR NOTE ON THE PLANS OR IN THE SPECIFICATIONS, THE STRICTEST PROVISION SHALL GOVERN.
3. THE STRUCTURAL DRAWINGS ARE FOR THE PLACEMENT OF THE PROJECT STRUCTURAL COMPONENTS ONLY. REQUIREMENTS MADE BY OSHA, DNR, AND ALL APPLICABLE SAFETY CODES ARE TO BE DETERMINED AND PROVIDED BY THE CONTRACTOR.
4. THE STRUCTURE IS DESIGNED TO BE SELF-SUPPORTING AND STABLE AFTER IT IS FULLY COMPLETED ACCORDING TO THE PLANS AND SPECIFICATIONS. IT IS THE CONTRACTOR'S SOLE RESPONSIBILITY TO DETERMINE CONSTRUCTION / ERECTION PROCEDURE AND SEQUENCE, AND TO ENSURE THE SAFETY OF THE STRUCTURE AND ITS COMPONENT PARTS DURING CONSTRUCTION. THIS INCLUDES PROVIDING TEMPORARY BRACING, SHORING, GUYS OR TIE-DOWNS. THESE TEMPORARY SUPPORTS SHALL REMAIN IN PLACE UNTIL ALL STRUCTURAL COMPONENTS ARE IN PLACE AND COMPLETED AS THE STRUCTURAL MEMBERS OR SYSTEMS ARE NOT SELF-BRACING UNTIL PERMANENTLY CONNECTED TO THE STRUCTURE.
5. THE ARCHITECT AND STRUCTURAL ENGINEER ASSUME NO LIABILITY FOR THE STRUCTURE DURING CONSTRUCTION. AS SUCH, THE MEANS AND METHODS OF CONSTRUCTION ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR(S).
6. USE OF THESE ENGINEERING DRAWINGS, PLANS OR DETAILS USED AS ERECTION PLANS OR SHOP DRAWINGS BY THE CONTRACTOR IS EXPRESSLY PROHIBITED. SUBMITTALS BEARING IMAGES ELECTRONICALLY COPIED FROM THE ENGINEERING DRAWINGS WILL BE REJECTED.

DEFERRED SUBMITTALS

COMPONENTS OF THE PROJECT DESIGNED BY THE CONTRACTOR'S SPECIALTY ENGINEER WHICH REQUIRE PRODUCT OR SYSTEM ENGINEERING, SEALED SHOP DRAWINGS AND CALCULATIONS TO BE SUBMITTED FOR REVIEW BY THE ENGINEER OF RECORD ARE AS FOLLOWS:

- COLD-FORMED METAL FRAMING
- CURTAIN WALL SYSTEMS

SHOP DRAWING SUBMITTALS

SHOP DRAWINGS, FABRICATION DETAILS, PRODUCT LITERATURE AND CERTIFICATES SHALL BE SUBMITTED BEARING THE CONTRACTOR'S REVIEW STAMP FOR THE FOLLOWING STRUCTURAL SYSTEMS. FAILURE OF THE CONTRACTOR TO STAMP SUBMITTAL PRIOR TO FORWARDING THEM TO THE DESIGN PROFESSIONAL(S) FOR REVIEW SHALL CONSTITUTE GROUNDS FOR REJECTION. USE OF THESE ENGINEERING DRAWINGS, PLANS OR DETAILS USED AS ERECTION PLANS OR SHOP DRAWINGS BY THE CONTRACTOR IS EXPRESSLY PROHIBITED. SUBMITTALS BEARING IMAGES ELECTRONICALLY COPIED FROM THE ENGINEERING DRAWINGS WILL BE REJECTED.

- CONCRETE COMPONENTS, MIX DESIGNS AND CONCRETE REINFORCEMENT
- MASONRY COMPONENTS AND REINFORCEMENT
- STRUCTURAL STEEL, OPEN WEB JOISTS, METAL DECK AND METAL FABRICATIONS
- COLD-FORMED METAL FRAMING
- CURTAIN WALL SYSTEMS

EXISTING CONDITIONS

VERIFY ALL DIMENSIONS AND CONDITIONS ASSUMED AS EXISTING (I.E. EXISTING MATERIALS; FOUNDATION SIZES AND CONFIGURATION, FRAMING MEMBER SIZES AND LOCATIONS; METHODS OF CONSTRUCTION; ETC.) AT THE SITE PRIOR TO CONSTRUCTION AND FABRICATION. IF DISCREPANCIES ARE FOUND, NOTIFY THE ARCHITECT PRIOR TO PROCEEDING.

SHORING

1. WHILE DETERMINING THE "MEANS AND METHODS" OF CONSTRUCTION, THE CONTRACTOR MAY DEEM SHORING TO BE NECESSARY TO PROVIDE A SAFE WORK PLACE. CONTRACTOR SHALL RETAIN THE SERVICES OF A PROFESSIONAL ENGINEER REGISTERED IN THE STATE OF MICHIGAN TO PREPARE ALL REQUIRED SHORING AND TEMPORARY BRACING DRAWINGS, INCLUDING THE CONTRACTOR'S PROPOSED SHORING PROCEDURE. SEALED SHORING DRAWINGS, CALCULATIONS AND SHORING PROCEDURE SHALL BE SUBMITTED FOR APPROVAL PRIOR TO COMMENCING WORK.
2. CONTRACTOR SHALL ESTABLISH A BENCHMARK TO DETERMINE AND MONITOR ALL ELEVATION CHANGES DURING THE PROCEDURE.
3. CONTRACTOR SHALL PROVIDE CRIBBING OR OTHER APPROVED MEANS TO LIMIT THE CONTACT PRESSURE TO THE EXISTING SLAB ON GRADE TO A MAXIMUM OF 1500 PSF.
4. SHORING SHALL BE PERFORMED USING CALIBRATED JACKS. JACK SPECIFICATIONS SHALL BE SUBMITTED FOR APPROVAL PRIOR TO COMMENCING WORK.
5. SHORING LOAD SHALL BE APPLIED TO THE STRUCTURE UNIFORMLY AND GRADUALLY, APPROXIMATELY 20% EVERY 30 MINUTES, TO TRANSMIT FORCES TO THE STRUCTURE IN A STATIC MANNER.
6. SHORING LOAD SHALL BE REMOVED FROM THE STRUCTURE UNIFORMLY AND GRADUALLY, APPROXIMATELY 20% EVERY 30 MINUTES, TO PREVENT THE APPLICATION OF IMPACT FORCES ON THE STRUCTURE.

MATERIAL LEGEND

IMAGE	DESCRIPTION
	CONCRETE MASONRY
	CONCRETE MASONRY GROUT SOLID
	SUB-BASE
	CONCRETE
	EARTH
	STEEL

SYMBOL LEGEND

IMAGE	DESCRIPTION

ABBREVIATIONS

ACI = AMERICAN CONCRETE INSTITUTE
AESS = ARCHITECTURALLY EXPOSED STRUCTURAL STEEL
AISC = AMERICAN INSTITUTE OF STEEL CONSTRUCTION
ARCH. = ARCHITECT OR ARCHITECTURAL
ASCE = AMERICAN SOCIETY OF CIVIL ENGINEERS
ASTM = AMERICAN STANDARDS FOR TESTING AND MATERIALS
AWS = AMERICAN WELDING SOCIETY
BM. = BEAM
B.O. = BOTTOM OF
BRG. = BEARING
CL. = CENTERLINE
DN. = DOWN
DNR = DEPARTMENT OF NATURAL RESOURCES
DWGS. = DRAWINGS
E.F. = EACH FACE
ELECT. = ELECTRICAL
ELEV. = ELEVATION
E.W. = EACH WAY
EX., EXIST. = EXISTING
EXP. = EXPANSION
F.F. = FINISH FLOOR
G.T. = GIRDER TRUSS
HI = HIGH
HORIZ. = HORIZONTAL
KIPS = KILOPOUNDS
L.L.V. = LONG LEG VERTICAL
L.L.H. = LONG LEG HORIZONTAL
LO = LOW
MAS. = MASONRY
MAX. = MAXIMUM
MBC = 2015 MICHIGAN BUILDING CODE
MECH. = MECHANICAL
MIN. = MINIMUM
MPC = MASS POURED CONCRETE
MPH = MILES PER HOUR
NCMA = NATIONAL CONCRETE MASONRY ASSOCIATION
O.C. = ON CENTER
OSHA = OCCUPATION SAFETY AND HEALTH ADMINISTRATION
PEMB = PRE - ENGINEERED METAL BUILDING
PERT = PRE - ENGINEERED ROOF TRUSS
PL, PLS = PLATE(S)
PSI = POUNDS PER SQUARE INCH
PSF = POUNDS PER SQUARE FOOT
SDI = STEEL DECK INSTITUTE
SF = SQUARE FEET
SIM. = SIMILAR
SJI = STEEL JOIST INSTITUTE
STRUCT. = STRUCTURAL
T.O. = TOP OF
T.O.S. = TOP OF STEEL UNDERSIDE OF SLAB OR ROOF DECK
TYP. = TYPICAL (SITUATION OCCURS REPEATEDLY)
U.N.O = UNLESS NOTED OTHERWISE
V., VERT. = VERTICAL
V.I.F. = VERIFY IN FIELD

STRUCTURAL CODE DATA

DESIGN CRITERIA PER 2015 MICHIGAN BUILDING CODE AND ASCE-16 LOAD STANDARD

1604.5 BUILDING OCCUPANCY:
CATEGORY III
1603.1.2 ROOF LIVE LOAD: MINIMUM = 20 PSF
1607.3 FLOOR LIVE LOAD: MINIMUM = 40 PSF
1603.1.3 ROOF SNOW LOAD:
GROUND SNOW LOAD Pg = 25 PSF
EXPOSURE = B
FLAT ROOF SNOW Pf =20 PSF
EXPOSURE FACTOR Ce = 1.00
IMPORTANCE FACTOR Is = 1.10
THER,AL FACTOR Ct = 1.00
1603.1.4 WIND LOAD:
BASIC EIND SPEED = 120 MPH
EXPOSURE = B
WIND IMPORTANCE FACTOR Iw = 1.00
INTERNAL PRESSURE COEFFICIENT = +/-0.18
ULTIMATE COMPONENTS AND CLADDING PRESSURES:
ROOF (50 S.F) = -24.3 PSF (ZONE 1)
ROOF (50 S.F) = -32.7 PSF (ZONE 2)
ROOF (50 S.F) = -39.3 PSF (ZONE 3)
WALL (50 S.F) = -23.2 PSF (ZONE 4)
WALL (50 S.F) = -26.7 PSF (ZONE 5)

1603.1.5 EARTHQUAKE DESIGN DATA:
SEISMIC IMPORTANCE FACTOR Ie = 1.00 MAPPED
SPECTRAL RESPONSE COEFFICIENTS: Ss = 0.103g
S1 = 0.046g
SITE CLASS = D
MAPPED SPECTRAL RESPONSE COEFFICIENTS: Sds = 0.109g
Sd1 = 0.073g
SEISMIC DESIGN CATEGORY B
BASIC SEISMIC FORCE RESISTING SYSTEMS
ORDINARY MASONRY SHEAR WALLS
SEISMIC RESPONSE COEFFICIENTS: Cs = 0.026
RESPONSE MODIFICATION FACTOR: R = 6.5
DESIGN BASE SHEAR = 0.026W
ANALYSIS PROCEDURE USED:
EQUIVALENT LATERAL FORCE PER ASCE 7, SECTION 12.8
1603.1.6 FLOOD DESIGN DATA:
THIS STRUCTURE IS NOT LOCATED IN A FLOOD HAZARDOUS AREA.
1603.1.7 SPECIAL LOADS:
REFER TO ROOF FRAMING PLANS FOR ROOFTOP EQUIPMENT LOADS.
1603.1.8 SYSTEMS AND COMPONENTS FOR SEISMIC RESISTANCE:
BY ITS SEISMIC DESIGN CATEGORY B, THIS STRUCTURE HAS NO REQUIREMENT FOR SYSTEMS AND COMPONENTS REQUIRING SPECIAL INSPECTIONS FOR SEISMIC RESISTANCE.

DRAWING INDEX

● ISSUED AS NEW OR REVISED
○ ISSUED FOR REFERENCE ONLY

LICENSE #: 6201043820.
LICENSE EXPIRES ON: 04/22/23
THE PROFESSIONAL ENGINEER SEAL COVERS ONLY THE SHEETS LISTED BELOW

SHEET	SHEET TITLE	CLIENT REVIEW 05-12-20	COST ESTIMATE 07-22-20	PERMIT 11-30-20
S0.00	GENERAL NOTES	●	●	●
S1.00	FOUNDATION & 1ST FLOOR FRAMING PLAN	●	●	●
S2.01	1ST FLOOR FRAMING PLAN			
S2.02	2ND FLOOR FRAMING PLAN	●	●	●
S2.03	3RD FLOOR FRAMING PLAN			
S2.04	4TH FLOOR FRAMING PLAN			
S3.01	ROOF FRAMING PLAN	●	●	●
S4.01	SHEAR WALL PLAN			
S4.02	SHEAR WALL PLAN			
S5.00	FOUNDATION DETAILS	●	●	●
S6.00	FRAMING DETAILS	●	●	●
S7.00	SPECIAL INSPECTION SHEET			

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Registration Seal:
11-30-20

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1588 ELM, DETROIT, MI, 48216

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GENERAL NOTES

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